R16

Code No: 133BD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, September - 2021
MATHEMATICS – IV
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, ETM, MMT, AE, MIE, PTM, CEE, MSNT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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Used Papers SEP 2021

1.a) Find the analytic function whose real part is $\left(x^2-y^2\right)e^{-y}\cos x - e^{-y}.2xy\sin x$.

b) Let *u* and *v* denote the real and imaginary components of the function *f* defined by means of the equations $f(z)=\begin{cases}\bar{z}^2/z & when\ z\neq 0\\ 0 & when\ z=0\end{cases}$. Verify whether the Cauchy–Riemann equations are satisfied or not at the origin $z=(0,0)$. [8+7]

2.a) Prove that if $w=f(z)=u+iv$ is analytic in a region R, then $dw/dz=\partial w/\partial x=-i(\partial w/\partial y)$

b) Evaluate $\oint_C (x+2y)dx+(y-2x)dy$ along:
i) the parabola $y=2x^2$ from (1, 2) to (2, 8)
ii) the straight lines from (1, 1) to (1, 8) and then from (1, 8) to (2, 8)
iii) the straight line from (1, 1) to (2, 8). [7+8]

3.a) Evaluate $\int_C \frac{e^z dz}{(z-1)^2}$ using cauchy's integral formula, where C is $|z|=2$.

b) Find the poles and the corresponding residues of each pole of $f(z)=\frac{(z-1)}{(z-2)^2(z-3)}$ [7+8]

4.a) Find the Maclaurin series and its radius of convergence $\frac{z+2}{1-z^2}$

b) Find the Laurent series that converges for $0<|z-z_0|<R$ and determine the precise region of convergence. Show details $\frac{z^2-3i}{(z-3)^2},\ z_0=3$. [7+8]

5.a) Find all the singularities in the finite plane and the corresponding residues.
$\frac{\cos z}{z^4}$

b) Evaluate (counterclockwise).
$\oint_C \frac{z-23}{z^2-4z-5}$ $C:|z-2-i|=3.2$ [8+7]

6. Evaluate the integral $\int_{-\infty}^{\infty}\frac{x}{8-x^3}dx$. [15]

7.a) Obtain the fourier series for $f(x) = x^2 \; in [-\pi, \pi]$.

b) Find the cosine transform of $\hat{f}_c(w)$ of $f(x) = \begin{cases} x \; if & 0 < x < 2 \\ 0 \; if & x > 2 \end{cases}$ [7+8]

8. Find the deflection $u(x,t)$ for the string of length L and $c^2 = 1$ when the initial velocity is zero and the initial deflection with k (say, 0.01) is $k \; sin \; 3\pi x$. [15]

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